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MILITARY SPECIFICATION

CONVERTER, LIQUID- TO GAS-OXYGEN, ARGON, AND NITROGEN; SKID-MOUNTED, 1,200 SCF PER HOUR CAPACITY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a skid-mounted, electric-motor-driven converter used in converting oxygen, argon, and nitrogen from liquid to gas, by means of a pump and a vaporizing unit, then charge the gas into cylinders through a manifold.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3655

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

W-R-550	- Rod, Ground (With Attachments).
CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
QQ-A-250/1	- Aluminum 1100, Plate and Sheet.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-T-5021	- Tests; Aircraft and Missile Welding Operators' Qualification.
MIL-S-12514	- Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-E-16298	- Electric Machines Have Rotating Parts and Associated Repair Parts: Packaging of.
MIL-C-52211	- Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of.

STANDARDS

FEDERAL

FED. STD. No. 66	- Steel: Chemical Composition and Hardenability.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Checks, 1/8 Pipe Threads, Steel, Type III.
MS35844	- Plug, Machine Thread, Magnetic: Drain.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B31.1 - Power Piping.
Y14.5 - Dimensioning and Tolerancing.
S1.4 - Specification For Sound Level Meters.
S1.11 - Specification For Octave, and Third Octave Band Filter Sets.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN WELDING SOCIETY, INC.

Welded Highway and Railway Bridges Design, Construction, and Repair.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

RUBBER MANUFACTURERS ASSOCIATION

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Description. The converter shall consist of a skid-mounted, high-pressure, liquid pump, driven by an electric motor through a belt or gear reduction train; a vaporizing unit; a cylinder charging manifold unit; and all necessary piping, valves, and gages required for functional completeness when the converter is operated in conjunction with liquefied atmospheric gas storage containers and gas cylinders. The converter shall convert liquefied oxygen, argon, and nitrogen to gas and charge it into cylinders at the pressure specified for each gas - 2200 psi for oxygen and argon, and 3500 psi for nitrogen.

3.2 Preproduction model. The contractor shall furnish a converter for examination and test within the time frame specified to prove, prior to starting production, that his production methods will produce converters that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified, the contractor shall furnish to the Government one or more converters for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be subject to all provisions of this specification (see 6.9).

3.4.1 Low-temperature metals. Steel conforming to FED. STD. No. 66, Type 304 or 304L, 316, or 321, or the 400 series steel, may be used where the contractor's standard practice is to insert a steel sleeve into a pump cylinder casting, or aluminum alloy, or copper alloys shall be used for parts subject to temperatures below minus 200° F.

3.4.2 Corrosion-resistant metals. All metallic surfaces of the converter that contacts high-purity oxygen shall be fabricated from corrosion-resistant material. Corrosion-resistant steel, except steel specified in 3.4.1, shall conform to FED. STD. No. 66, Type 301 through 348, as applicable. Copper, brass, bronze, silver, beryllium-copper-nickel alloys may be used in lieu of corrosion-resistant steel. Metal parts, including bolts, nuts, screws, doorhinges, locks, latches, hasps, pins, fasteners, and washers shall be corrosion-resistant or zinc or cadmium coated to commercial quality.

3.4.3 Nonmetallic materials. Nonmetallic materials adversely affected by continued use with gaseous or liquid oxygen shall not be used. The non-metallic materials used shall show no reaction when tested as specified in 4.6.2.3. Tetrafluoroethylene material shall be acceptable without testing.

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3.5 Environmental conditions.

3.5.1 Operating temperature requirements. The converter shall perform as specified herein in any ambient temperature from plus 120° F to minus 25° F.

3.5.2 Storage temperature requirements. The converter shall not be damaged by storage under the following conditions:

Air temperature:

High - plus 155° F. Time period, 4 hours daily.
Low - minus 65° F. Time period, 12 hours. No solar radiation.

3.5.3 Rain. The converter shall withstand rain at a rate of 3 inches per hour impinging at angles varying from the vertical to 45 degrees when directed simultaneously at the top and at all four sides without leaking into the interior. Operation of the converter shall not be affected by the rain.

3.6 Safety. All exposed parts which are subject to extreme operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions. Drain valves and pressure-relief devices shall be arranged so that gas or liquid does not impinge on the converter or on operating personnel. Warnings shall be mounted on or near components containing hidden hazards.

3.6.1 Pressure-relief devices. Pressure-relief devices shall be provided in any line or component that can be isolated by closing valves, or that would allow dangerous pressure buildup to develop.

3.6.2 Emergency equipment. The converter shall have provisions for the location of emergency equipment, including at least one 5-pound carbon dioxide fire extinguisher located inside the pump and control cabinet, 3.15.

3.6.3 Sound level. The sound level of the converter shall be in accordance with MIL-STD-1474, Category D for steady-state noise, when tested in accordance with 4.6.2.8.

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3.7 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 240 hours when the converter is tested as specified in 4.6.2.6.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.4), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.5.

3.8.2 Maintenance ratio. The converter shall have a maintenance ratio of not more than 0.06. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8.3 Scheduled maintenance. The converter shall not require scheduled maintenance more often than every 12 hours of operation when tested as specified in 4.6.2.5.1. Scheduled maintenance including normal scheduled adjustments, refilling the liquid container, and servicing the converter shall require not more than a 1/2-hour period.

3.8.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.9 Performance.

3.9.1 Capacity. The converter, when operating in ambient conditions of plus 100° F and 85 percent relative humidity, shall pump liquid from a liquid storage tank pressurized from 20 psi to 100 psi, shall vaporize the liquid, warm the resulting vapor to within 20° F of ambient temperature, and charge the vapor into gas storage cylinders at a rate of not less than 1200 standard cubic feet per hour (see 6.4) to a cylinder pressure of 3500 psig, at 70° F. (The pump shall be capable of pressuring up to 4000 psig to permit filling of nitrogen cylinders at temperature, above 70° F.

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3.9.2 Operating time. The converter shall operate continuously under the conditions at the pressure specified in 3.9.1 for periods of not less than 12-hours without requiring shutdown for defrosting, servicing, maintenance, or repairs.

3.10 Configuration and weight.

3.10.1 Configuration. The converter dimensions, including all accessories when attached for transport, shall not exceed 62 inches in width, 72 inches in length and 68 inches in height.

3.10.2 Weight. The converter when complete with all accessories, charging manifold and cylinder holding rack, liquid product hose, vapor hose, gas product hose, charging hose with adapters, and fire extinguisher shall weigh not more than 1800 pounds.

3.11 Fabrication. The converter shall conform to the ASME Boiler and Pressure Vessel Code or to ANSI B31.1, as applicable.

3.11.1 Skid base. The converter shall be mounted on two longitudinal skids from structural steel. A 1/2-inch hole shall be provided through the skid surface at each end of each skid for bolting to a crate floor. A sheet steel closure shall be provided between the skids. The skids shall be braced and gusseted to prevent sagging or distortion of the converter. Both ends of each skid frame shall be reinforced and shall be provided with eye holes, clevices, or other means to permit pulling the converter over the ground. The eyes or clevices may also be used as lifting or tiedown attachments when they meet the requirements of 3.16 or 3.17. The devices furnished to permit pulling shall have an opening of not less than 3-inch inside diameter. The bottom tread of the skids shall be curved or turned up at each end to prevent snubbing when pulled over uneven surfaces. The bottom tread of each skid shall have a 1/2-inch hole drilled at each end for bolting the converter to a crate floor. The closure on the skid platform shall be provided with drains to prevent the accumulation of water. Provisions shall be made for handling the converter with a forklift. Tubes shall be provided in the base of the skid frame to receive the forklift tines and prevent possible damage to the converter or its components while being handled.

3.11.2 Pump. The pump shall be a single-cylinder, reciprocating-plunger- or piston-type unit that will receive liquid at 20 to 100 psi above atmospheric pressure and deliver the liquid to the vaporizer at a pressure of 4000 psig. The pump shall be so constructed as to facilitate disassembly, maintenance, repairs, and reassembly of all working parts. When replaceable cylinder sleeves are furnished, they may be fabricated from high-abrasive-resistant steels such as Alloy 440C in accordance with FED. STD. No. 66.

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The pump shall be designed to be self-sealing from the atmosphere when not in use. When operating as specified herein, the pump shall not freeze out ice from the surrounding air on the packing or plunger rod to the extent that it will interfere with continuous operation of the pump. When adjustable plunger-rod packing glands are furnished, a means shall be provided for making necessary adjustments while the pump is in operation. No liquid or vapor product shall be blown to the atmosphere during operation. A return line and connections shall be provided for returning the flash-off gas to a storage tank.

3.11.2.1 Low-pressure product filter. A porous-element-type filter shall be incorporated into the pump suction line to remove metal and dirt particles (in excess of 100 microns in size) before they enter the liquid pump. The filter shall be located where it is readily accessible for removal when cleaning and replacing the filter element.

3.11.2.2 High-pressure product filter. The filter as specified in 3.11.2.3 shall be provided in the pump discharge line to remove particles entering with the liquid steam.

3.11.2.3 Check valve. A check valve shall be installed in the product-pump discharge line up stream from the high-pressure product filter to prevent back-flow of the product to the pump assembly.

3.11.2.4 Pump compatibility with liquid tank. The converter shall have the compatibility of receiving feed liquid from an unrefrigerated cryogenics storage tank having a liquid capacity of no less than 100 gallons and no more than 400 gallons. All pump flash-off gas shall be returned to the storage tank. The pump shall operate continuously at capacity without interruption, until it has extracted not less than 90 percent of the storage tank.

3.11.3 Motor and speed reduction assembly. A motor and speed reduction assembly shall be furnished to drive the liquid pump. The motor shall conform to CC-M-1807 Type II, Class 11 or 13, Style B2, continuous duty, 1800 rpm, 220/440 volts, 60-cycle. All electrical wiring shall be in accordance with NFPA No. 70 for hazardous locations. The motor shall be provided with means for operating on either 220 or 440 volts, and shall be initially wired for 440 volt service with necessary instructions for converting to 220 volts. A directly connected, completely enclosed, gear- or belt-type speed reduction assembly with a self-contained lubrication system shall be provided to adapt the electric motor shaft speed to that speed required for operation of the liquid-product pump under full load at not more than 100 rpm. The electric motor and speed reduction assembly shall be installed so that the motor output shaft is correctly alined with the

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liquid pump drive shaft or with the eccentric drive shaft of the product pump for direct drive through a transmission coupling. The motor and pumping unit shall be held stationary by bolts extending through the skid base floor. A set of positioning screws shall be attached to the skids for making and retaining horizontal adjustments between the electric motor and pump on units equipped and driven with a drive belt(s).

3.11.4 Cabinet enclosure. A cabinet having two compartments shall be provided. One compartment shall be provided for enclosing the working and control equipment, and one compartment for storage. The cabinet shall be provided with internal supports for securing the enclosed equipment in its correct position under all conditions of service specified herein. Provisions shall be incorporated to insure protection from damage to components resulting from temperature variation and stresses occurring during operation and transport. The cabinet shall be of sufficient size to enclose all the converter components and their insulation, and provide separated storage space for the service hose, charging manifold, power supply cable, the ground rod, accessories for the cylinder charging rack, charging lead w/adapters, and fire extinguisher rack. The cabinet shall be provided with doors suitably located to permit access to all enclosed equipment for operation and maintenance. Means shall be provided to secure the doors in the open and closed positions. Door handles shall be provided and shall have means for accommodating a 3/8-inch padlock for locking. When an insulated pump compartment is furnished inside the cabinet, access panels and door joints shall be provided with gaskets to prevent heat travel into the insulated compartments. Airtight insulated compartments shall be provided with a relief to vent the internal pressure when it reaches 5 psi.

3.11.4.1 Insulation. All components in the converter cabinet that operate at low temperatures, where heat transfer is not desirable, shall be insulated. The insulation shall be of a type that is not flammable and does not absorb moisture. Formed insulation shall be used where possible. The insulation shall be removable and replaceable to permit cleaning and repair. Band-type clamps shall be used for retaining formed insulation in place.

3.11.5 Vaporizer. The vaporizer shall receive the liquefied atmospheric gas from the pump, vaporize it, and cause a temperature rise to within 20° F of the ambient temperature. The vaporizer shall be of the fin and tube type assembled in parallel and constructed from corrosion-resistant steel tubing with heat dissipation aluminum fins. The vaporizing coils shall be enclosed in an expanded metal cover which will permit free flow of air at ambient temperature. Brackets shall be provided to affix the coils to the enclosure and to maintain a fixed space between the coil sections of the tube and fin

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assembly. The floor of the vaporization coil enclosure shall be constructed from extruded metal of sufficient strength to prevent damage to the coils during handling and while in transit.

3.11.6 Product transfer lines.

3.11.6.1 Liquid transfer lines. Two 12-foot-long, 1/2-inch-inside-diameter, flexible corrosion-resistant steel hoses with 2-1/2-inch-outside-diameter corrosion-resistant steel braid shall be furnished. Insulation of the liquid transfer lines may be by fiber glass pack or by vacuum shell. Lines insulated with fiber glass shall have the insulation placed between the 1/2-inch inner steel hose and the 2-1/2-inch outer corrosion-resistant steel braid. The outer corrosion resistant steel braid shall be held in place by wooden spacers placed approximately every 2 feet on the outside of the inner hose. The end of the hose assembly shall be sealed to prevent leakage of insulation. The line end fittings shall consist of steel nipples which shall be welded to the inner hose and the outer shell of the line. Each line end shall be terminated in a 1/2-inch NPT externally threaded nipple. The vacuum lines shall be evacuated by use of a standard vacuum pump to 5 microns. When vacuum insulated lines are furnished a special vacuum adapter valve shall be furnished with each converter for reevacuating the hose while in field use. The transfer lines shall have a burst pressure of 2000 psi. Provisions shall be made for onboard stowing of the transfer line assemblies when not in use (see 3.14.4).

3.11.6.2 Vapor transfer line. One tube assembly shall be furnished for connecting the vaporizer to the charging manifold. The assembly shall withstand the pressure specified herein. Construction shall be of flexible copper or corrosion-resistant steel tubing not less than 12 feet long, with a number 540 connection on each end in accordance with Handbook H28. An attached wrench shall be provided on one end of the flexible line.

3.11.6.3 Couplings and adapters. Both ends of the liquid and gas lines shall be fitted with a quick-disconnect coupling assembly. Using an appropriate pipe bushing-reducer a 1-1/2 inch cryogenic coupling assembly (Future-Craft Corp. No. 00255/550043-15-000) or equal (female half) shall be attached to the line ends.

Note: The male half of each coupling shall be provided for adapting to the liquid service tank. Provisions shall be made for protecting the end connections from damage and dirt when the line is not in use. Retention chains and bands shall be provided for each protective cover.

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3.11.7 Charging manifold assembly. The charging manifold shall provide for the delivery of gas from the vaporizer coils into compressed-gas storage cylinders at the pressures and temperatures specified herein. The charging manifold shall be portable, and shall be in two sections, each section shall be equipped to service four gas cylinders. Check valves shall be provided in the manifold assembly to prevent backflow from one section after the cylinders have been filled and the distribution valve on the other section of the manifold assembly is opened. One manual distribution valve shall be provided on each half-section of the charging manifold assembly to handle 4000 psi pure nitrogen vapors. Each distribution valve, when closed, shall positively isolate its half-section of the charging manifold assembly. Each charging station shall be equipped with a manually operated shutoff valve with valve outlet No. 541 (see NBS Handbook H28) having a pressure rating of not less than 4000 psi. When closed, the valves shall positively isolate the cylinder being charged from the manifold section. A vacuum valve shall be furnished for each section of the charging manifold for connecting to the vacuum pump (see 3.11.9). A pressure discharge valve shall be furnished for each section of the charging manifold. Each charging-station valve shall be provided with one flexible charging lead having fittings for connecting to oxygen cylinder valves with valve outlet No. 541 (see NBS Handbook H28), one flexible charging lead having fittings for connecting to water-pumped nitrogen valve outlet No. 581 (see NBS Handbook H28), and one flexible charging lead having fittings for connecting to medical oxygen cylinder valves with outlet No. 871 (see NBS Handbook H28). The flexible charging leads shall be at least 24 inches in length and shall be constructed so they can be changed without special tools. The flexible leads shall be at least 24 inches in length and shall be so constructed that they can be changed without special tools. Each section of the charging manifold shall be mounted on a table for its support. Each charging manifold table shall be provided with cutouts and latch chains at each charging station for securing the cylinders in place, in the upright position, and to prevent movement or falling during the connecting and charging operations. The table shall be in two sections each 32 inches wide and 36 inches long. Each table shall have detachable legs of a height so that the charging manifold will not be lower than 4 to 6 inches below the shoulder of a standard 9 x 51-inch cylinder. Provisions shall be made for disassembling the manifold tables for storage and transit. The table legs and bracing shall be stored with the manifold in the converter cabinet. The table tops shall be stored and secured to the converter top. Check valves shall be provided in the manifold assembly to prevent backflow from one section after the cylinders have been filled and the distribution valve on the other section if the manifold assembly is opened. A pressure gage as specified in 3.11.8.2 shall be provided on the manifold assembly to indicate the pressure at which the cylinders are being charged. One pressure gage shall be mounted on each section of

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the manifold assembly. Two purging valves shall be provided for venting the charging manifold and leads of air or the gaseous product as applicable, before and after charging the cylinders. The purging valves shall have the same pressure rating as the charging manifold and shall be manually operated. They shall be located on opposite ends of the charging manifold at the end of each section serving four charging stations.

3.11.8 Controls and instruments. The converter shall be provided with all the controls and instruments necessary for use during continuous operation at any rate of production. All controls and instruments shall be centrally located inside the cabinet and arranged so the operator has access to the controls and a view of the instruments of both the converter and charging manifold during operation. All valves, gages, controls, and indicators shall be identified by securely attached plates in accordance with 3.19.2. All gages shall have a 4-1/2 inch dial, covered with a safety glass face and banded with a ring constructed from corrosion resistant material. The normal instrument operating reading range during operation shall be between one-fourth and three-fourth of the full reading scale. The pressure and temperature gage lines shall be grouped in an orderly manner, bonded and secured, and equipped with a vibration damping device. A shutoff and gas vent valve shall be provided for each pressure gage. Pressure gages used for oxygen service shall be marked in red with the following warning: "For oxygen service - use no oil." Pressure gages subject to pressures in excess of 250 psi shall be provided with safety backs.

3.11.8.1 Product temperature gage. A product temperature gage shall be provided to indicate the temperature of the product at the outlet of the vaporizer. The gage shall be accurate to within plus or minus 2 degrees.

3.11.8.2 Pump inlet pressure gage. A pressure gage of the Bourdon tube, dial-indicating-type shall be provided to indicate the liquid pressure at the inlet side of the pump. The scale shall be from 0 to not more than 200 psig with each 20 psig increment numbered. The gage shall be located on the instrument panel and connected at the pump inlet.

3.11.8.3 Pump discharge pressure gage. A pressure gage of the Bourdon tube, dial-indicating-type shall be provided to indicate the pressures at which the liquefied gas is pumped to the vaporizer. The normal discharge pressure shall be not less than 4000 psig. The gage scale shall cover a range from 0 pressure by 500 psig increments to not less than 4,000 psig. The gage scale shall be marked in bold numbers at each 500 psig increment on the scale. Each 500 psig increment shall be numbered at each 100 psig mark in number succession. The gage shall be mounted on the instrument panel and connected into the discharge line between the pump and vaporizer.

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3.11.8.4 Discharge safety valve. A pump discharge safety valve having a blowdown of not less than 5 percent of the preset pressure shall be furnished.

3.11.8.5 Vent valve. A vent valve shall be provided to vent all high-pressure piping during startup and after operation. The vent valve shall be a manually-operated, brass, angle valve.

3.11.8.6 Pump drain valve. A pump drain valve shall be provided. The valve shall be a manually-operated, brass, globe valve. The pump drain valve shall be at a lower elevation than that of the pump casting.

3.11.8.7 Liquid inlet connection and valve. The inlet to the converter shall be a connector with 1/2-inch 14-NPT internal threads located for connection to the converter. Control of the flow of liquefied atmospheric gas to the pump from an auxiliary cryogenic tank shall be provided by incorporating an inlet valve (cold valve) on the plunger end of the pump. The valve shall be a manually-operated, flange-mounted, long-stem type, packed, and shall be suitable for liquid oxygen service.

3.11.9 Vacuum pump. A vacuum pump shall be furnished, capable of evacuating four, 340 cubic foot capacity cylinders, to a pressure of 3 inches of mercury (Hg) or less, in 30 minutes or less, from one atmosphere. Two flexible lines shall be provided for connection between the vacuum pump and the two vacuum shutoff valves on the charging manifold (see 3.11.7). The pump shall be driven by an electric motor operating on the electrical power specified in 3.11.3. The electric motor shall be in accordance with 3.11.3. Power shall be transmitted between the electric motor and the vacuum pump by V-belts, a gear reduction, or a direct-drive system. When used, V-belts shall be in accordance with Rubber Manufacturers Association (RMA). Gages, as specified in 3.11.8 shall be provided as necessary for safe operation.

3.12 Electrical system.

3.12.1 Power-supply cable. A power-supply cable conforming to MIL-C-3432, Type CO-04 HLF (4/12 0685), 100 feet long, shall be furnished. The cable shall have a capacity to carry not less than 220/440 volts a.c. The cable shall be portable and shall have three conductors and ground. The power conductors shall be color coded white, black, and red. The ground wire shall be green. One end of the cable shall be permanently attached to the converter. The other end shall be provided with bared conductors for connection to solderless terminals. Provisions shall be made for stowing the cable inside the converter cabinet when not in use or during storage and transport.

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3.12.2 Starting switch. A starting switch with thermal overload protection, enclosed in an explosion-proof cast metal enclosure of sufficient size to service the pump motor (see 3.11.3) shall be furnished. The enclosure containing the starter and overload shall be mounted inside the motor-pump cabinet in a location that will permit access to all controls. The starter shall conform to MIL-S-12514, Type I, Class 2, size as applicable. The switch shall be marked to indicate the "ON" and "OFF" positions.

3.12.3 Grounding. The motor frame and all electrical equipment enclosures shall be grounded to the skid with bar copper conductors of No. 6 AWG wire. The skid shall be grounded to a driven ground-rod assembly through a single conductor cable conforming to MIL-C-3432, Type CO-01HLF(1/6)0510, connected to a solderless terminal fastened to the skid. The system neutral shall be grounded at each neutral bus.

3.12.4 Ground rod. A ground-rod assembly conforming to W-R-550, Type III, Class B shall be furnished except the ground wire furnished shall be 15 feet long. Provisions shall be made for securing the ground-rod assembly, clamps, and wires inside the converter storage cabinet when they are not in use.

3.13 Electromagnetic interference. The electromagnetic interference emission of the converter shall conform to MIL-STD-461, Class IIB.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14.1 Lubricants. The converter shall operate as specified herein when lubricated with military lubricants (see 6.5).

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.14.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine

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the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the converter is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.14.5 Grease lubrication. Unless otherwise specified herein all grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

Halocarbon Oil Grease Type - 1121E NSN9150-00-935-4185 Oxygen Compatible Lube.

The converter shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because Commercial greases are not to be mixed with oxygen compatible greases. Each component shall be tagged in a conspicuous place to indicate which Commercial grease has been used.

3.14.6 Oxygen compatibility. In those areas of the converter where contact with oxygen is possible, oxygen compatible lubricants shall be used.

3.15 Fire extinguisher. Provisions shall be made for installing one 5-pound carbon dioxide fire extinguisher with extinguisher bracket which will be furnished by the Government (see 3.25). Space shall be provided inside the pump cabinets where it is accessible to the operator.

3.16 Lifting attachments. Permanently affixed attachments that enable the converter to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments conform to the requirements specified in 3.17.

3.17 Tiedown attachments. Permanently affixed attachments that permit fastening the converter to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall conform to MIL-STD-209. Attachments shall be rounded and beveled to prevent cutting of the tiedown material.

3.18 Accessory box. The converter shall be provided with an accessory box made of metal not less than 0.073 inch in nominal thickness and not less than 1 cubic foot in volume. The accessory box shall have a hinged

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lid and a trunk drawbolt of a type that will keep the lid closed when subjected to vibration. The accessory box shall be securely fastened to the converter in a protected position. The accessory box shall be of sufficient size to hold the charging leads 3.11.7 and all adapters and accessories in 3.24.

3.19 Identification and marking.

3.19.1 Identification plate. The converter shall be identified in accordance with MIL-STD-130. The marking shall be applied to the converter on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.19.2 Instruction plates. Each converter shall be provided with instruction plates or diagrams including warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the converter. Instruction plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. The instruction plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.19.3 Warning marking. The following warning shall be permanently affixed in a conspicuous place on the converter pump unit and on the charging manifold:

WARNING

DO NOT PERMIT OIL, GREASE, GREASY GLOVES

OR RAGS, OR OTHER COMBUSTIBLE MATERIAL

TO COME IN CONTACT WITH ANY PART OF THE

CONVERTER EXPOSED TO OXYGEN.

CLEANING

ALL FITTINGS AND EQUIPMENT

FOR OXYGEN SERVICE SHALL BE CLEANED WITH

A SUITABLE OXYGEN COMPATIBLE SOLVENT.

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3.19.4 Flow diagram. Each converter shall be provided with a flow diagram visible to the operator during operation. The direction of liquid flow for filling or emptying the converter shall be indicated by arrows. Each valve shall be identified by a number or letter and keyed to a legend identifying each valve as to its purpose. The plate shall be as specified in 3.19.

3.19.5 Wiring diagram plates. Electrical wiring schematic diagrams for the converter, electrical circuits shall be affixed near the controls in the machinery compartment by screws, bolts, or rivets. The plates shall be of a size to be clearly readable and shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material.

3.19.6 Operating instructions. Operating instructions shall be furnished and shall be permanently affixed by screws, bolts, or rivets near the control panel in the machinery compartment. Instructions shall be clear and concise and shall contain sufficient information to enable operation without damage to the plant or injury to personnel. The operating instructions shall be applied to plates conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material.

3.19.7 Valve marking. All control valves and safety valves shall be identified by a nameplate on or near the valve indicating a flow-control number. The nameplate information shall be applied to plates conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets.

3.20 Contact surfaces of dissimilar metals. Contact between dissimilar metals shall be avoided when practicable. The definition of dissimilar metals shall conform to MIL-STD-889. When such contacts do occur, they shall be separated by the use of joint fillers composed of zinc chromate, nonconductive aluminum powder, inert filler, oils, and thinner, and by the use of membrane materials. Membrane materials shall be of elastomer stripping not less than 1/32-inch thickness.

3.21 Treatment and painting. The portions of the converter and its components and parts normally painted in accordance with the contractor's commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, semigloss, Color No. 24087, when treatment and painting is required other than as specified (see 6.2).

3.22 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

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3.23 Repair parts and maintenance tools. Repair parts and other maintenance tools as are specified shall be furnished. Maintenance tools shall be securely affixed within the toolbox (see 6.2).

3.24 Items to be furnished with the converter. No shipment of converters shall be made unless repair parts, maintenance tools, operating tools, technical publications, and accessories are included with the shipment or unless approval for shipment of the converter without such items has been received from the contracting officer (see 6.8).

3.25 Government-furnished property. When specified, the following property in the quantities indicated will be furnished by the Government and mounted by the contractor (see 6.2):

Item No.	Description	Identification	Quantity for each plant
1	Fire extinguisher, CO, 5 lb.	NSN 4210-00-270-4512	1
2	Brackets, fire extinguisher	NSN 4210-00-268-9729	1

3.26 Workmanship.

3.26.1 Metal fabrication. Metals used in the fabrication of the converter shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting of ferrous metals may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Aluminum alloys up to 1/2 inch in thickness may be cut by shearing; above 1/2 inch in thickness, alloys shall be cut by common or high-speed saws. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.26.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided where necessary and all bolts, nuts, and screws be tight. Self-tapping sheet metal screws shall not be used.

3.26.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.26.3.1 Aluminum bolt and rivet holes. Bolt and rivet holes in aluminum alloys shall be made by any one of the following methods:

(a) Holes may be subpunched or subdrilled undersize and reamed to finish size upon assembly.
(b) Holes may be drilled to size at assembly.
(c) Holes may be drilled to size prior to assembly with accurately mated steel drilling jigs. The jigs shall be equipped with hardened steel bushings to insure the correct alinement and location of the holes. All burrs and rough edges shall be removed from the holes prior to final assembly.

3.26.4 Welders and welding.

3.26.4.1 Welders. Before assigning any welder to manual welding work on pressure systems covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes or in MIL-T-5021 for the type of welding operation to be performed, and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME.

3.26.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Under proof and service loading, welds shall develop strength and transmit stress to the parts connected without failure.

3.26.4.3 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein. Finished contact and bearing surfaces shall be in accordance with ANSI Y14.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification if inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The converter shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The converter shall be tested as specified in 4.6.2 through 4.6.2.12, except 4.6.2.6. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3) one or more initial production converters will be selected at random by the Government from the converters being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.6, 4.6.2.7, and 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production converter shall not exclude the remaining converters from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production converter to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production converters and shall be cause for refusal by the Government to continue acceptance of production converters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on converters previously accepted

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and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all converters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each converter shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each converter shall be subjected to the tests specified in 4.6.2.2.1 and 4.6.2.14. Failure of either test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. Each converter shall be examined for the following defects as specified in 4.5.1:

Number	Defect	Requirement Paragraph
Major		
101.	Parts or components missing.	3.1
102.	Material not as specified.	3.4
103.	Low-temperature metals not as specified.	3.4.1
104.	Corrosion-resistant metals not as specified.	3.4.2
105.	Nonmetallic materials not as specified.	3.4.3
106.	Pressure-relief devices not furnished as specified.	3.6.1
107.	No provisions for the location of emergency equipment as specified.	3.6.2
108.	Configuration not as specified.	3.10.1

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109.	Weight not as specified.	3.10.2
110.	Fabrication not in accordance with the ASME Boiler and Pressure Vessel Code or ANSI B31.1 as specified.	3.11
111.	Skid base not as specified.	3.11.1
112.	Pump not as specified.	3.11.2
113.	Low pressure product filter not as specified.	3.11.2.1
114.	High pressure product filter not as specified.	3.11.2.2
115.	Check valve not as specified.	3.11.2.3
116.	Pump not compatible with liquid tank as specified.	3.11.2.4
117.	Motor and speed reduction assembly not as specified.	3.11.3
118.	Cabinet enclosure not as specified.	3.11.4
119.	Insulation not as specified.	3.11.4.1
120.	Vaporizer not as specified.	3.11.5
121.	Liquid transfer lines not as specified.	3.11.6.1
122.	Vapor transfer line not as specified.	3.11.6.2
123.	Couplings and adapters not as specified.	3.11.6.3
124.	Charging manifold assembly not as specified.	3.11.7
125.	Controls and instruments not as specified.	3.11.8
126.	Product temperature gage not as specified.	3.11.8.1
127.	Pump inlet pressure gage not as specified.	3.11.8.2

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128.	Pump discharge pressure gage not as specified.	3.11.8.3
129.	Discharge safety valve not as specified.	3.11.8.4
130.	Vent valve not as specified.	3.11.8.5
131.	Pump drain valve not as specified.	3.11.8.6
132.	Liquid inlet connection and valve not as specified.	3.11.8.7
133.	Power-supply cable not as specified.	3.12.1
134.	Starting switch not as specified	3.12.2
135.	Grounding not as specified.	3.12.3
136.	Ground rod not as specified.	3.12.4
137.	Lubrication not as specified.	3.14
138.	Lubricants not as specified.	3.14.1
139.	Lubrication fittings not as specified.	3.14.2
140.	Pressure release device not as specified.	3.14.3
141.	Filling, draining, and checking provisions not as specified.	3.14.4
142.	Grease lubrication furnished not as specified.	3.14.5
143.	Instruction plates not as specified.	3.19.2
144.	Warning marking not as specified.	3.19.3
145.	Flow diagram not as specified.	3.19.4
146.	Wiring diagram plates not as specified.	3.19.5

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147.	Operating instructions not furnished as specified.	3.19.6
148.	Valve marking not as specified.	3.19.7
149.	Contact surfaces of dissimilar metals not handled as specified.	3.20
150.	Metal fabrication not as specified.	3.26.1
152.	Riveted connections not as specified.	3.26.3
153.	Aluminum bolt and rivet holes not as specified.	3.26.3.1
154.	Welders not qualified as specified.	3.26.4.1
155.	Welding not as specified.	3.26.4.2
156.	Machine work not as specified.	3.26.4.3
Minor		
201.	Fire extinguisher furnished not as specified.	3.15
202.	Accessory box not as specified.	3.18
203.	Identification plate not as specified.	3.19.1
204.	Treatment and painting not as specified.	3.21
205.	Technical publications not as specified.	3.22
206.	Repair parts and maintenance tools not as specified.	3.23

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified, all tests shall be performed without shelter and at the climatic conditions existing at the place of the test. Preproduction tests shall be conducted in the order in which they are listed. Necessary safety precautions shall be exercised in

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handling liquid oxygen and high-pressure gaseous oxygen in all operating and environmental conditions encountered. Prior to test, the converter shall be lubricated with oils and greases as specified in 3.14. Oil shall be those designated for use in the ambient temperature at the place of test.

4.6.2.2 Preassembly.

4.6.2.2.1 Pressure. Prior to assembly, components subject to pressure shall be tested in accordance with ASME Boiler and Pressure Vessel Code. Piping shall be tested in accordance with ANSI B31.1 using fluids compatible with oxygen in the liquid or gaseous state. Inability of any component to meet the requirements of the ASME Boiler and Pressure Vessel Code, or piping to meet requirements of ANSI shall constitute failure of this test.

4.6.2.2.2 Safety valves. Each safety valve shall be tested in accordance with ASME Boiler and Pressure Vessel Code. Inability of the safety valves to meet the requirements of the code shall constitute failure of this test.

4.6.2.3 Oxygen compatibility.

4.6.2.3.1 Contact with oxygen. Approximately 0.5 gram of insulating material of the type used in the converter shall be placed in a reaction chamber. The reaction chamber shall be heated to a temperature of plus 302° F and maintained at that temperature for 1 hour while under constant oxygen pressure of 2000 psig. The temperature shall be recorded at 5-minute intervals, and a temperature rise of 77° F or greater shall be accepted as an indication of a reaction. After the expiration of the 1-hour period, pressure on the chamber shall be released; after cooling, the chamber shall be opened for examination of the material. Three separate tests shall be conducted for each material. Discoloration or any other change in appearance of any specimen of material shall be accepted as evidence of combustion and shall constitute failure of this test.

4.6.2.3.2 Contact with oxygen under surge-pressure conditions. A reaction chamber shall be charged with 0.5 gram of insulating material and heated in an oven to plus 302° F. After the reaction chamber has been held at a temperature of plus 302° F for 10 minutes, an oxygen pressure of 2000 psig shall be applied suddenly. The temperature and pressure shall be maintained at a constant level for 5 minutes; at the end of this interval, the pressure shall be released and reapplied suddenly. A sudden temperature rise of more than 77° F during application of oxygen pressure, or a change in appearance of the specimen after test shall be considered evidence of combustion and shall constitute failure of this test.

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4.6.2.4 Impact sensitivity.

4.6.2.4.1 Apparatus. The impact sensitivity tester used for this test shall be similar to that shown in Figure 1. The plunger shall transmit a calculated impact energy of 60 foot-pounds plus 3 minus 0 foot-pounds to the test specimen held in an aluminum cup placed on the bottom plate (anvil) of the tester. The test cup shall be made of 1100 aluminum alloy, 1/16 inch thick conforming to QQ-A-250/1, and shall be approximately 3/4 inch inside diameter by 7/8 inch deep. The test shall be performed in a controlled atmosphere room. A protected operator's window and control panel shall be provided.

4.6.2.4.2 Test specimens. Test specimens shall consist of a specimen of each type of combustionable material used in the fabrication of the converter that will come in contact with oxygen, liquid or gas.

4.6.2.4.3 Test procedure. A test cup shall be used one time and discarded. The specimen, approximately 0.7 gram of material, shall be smeared over the bottom of the test cup. The cup and specimen shall then be precooled by dipping in liquid oxygen before transferring to the tester and topped by liquid oxygen just prior to impact. Repeat procedure for each type of material.

4.6.2.4.4 Failure criteria. Evidence of a reaction as indicated by an audible report, a visible flash, discoloration, or other evidence of burning shall constitute failure of this test.

4.6.2.5 Performance.

4.6.2.5.1 Scheduled maintenance. While performing the capacity test, one 1/2-hour period will be permitted for normal scheduled adjustments, refilling the liquid container, and servicing as specified in 3.8. Adjustments of the piston stroke will be permitted to regulate production. Interruptions in operation in excess of that specified in 3.8.3 shall constitute failure of this test.

4.6.2.5.2 Endurance. The converter shall be operated a total of 500 hours using 440 volt current as specified in 3.11.3. Except for servicing as specified in 4.6.2.5.1 operation shall be continuous. All operating time including maintenance and service time shall be logged. The entire operational run shall be accomplished under the conditions specified in 3.9. Leakage other than leakage which can be corrected by tightening valves or connections, nonconformance to 3.9.1 and 3.9.2, major repairs, wear of parts to the extent they are replaced, or failure to maintain capacity at 3500 psi shall constitute failure of this test.

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4.6.2.6 Reliability. Using the MTBF specified in 3.7, the initial production model converter shall be tested as specified in 4.6.2.5 with "accept" and "reject" criteria in accordance with MIL-STD-781, Test II. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting time excluded) by adjustment or repair, using controls, on-equipment tools or parts, and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
(b) Damage to the converter from continued operation.
(c) Safety hazards to personnel.
(d) Secondary or dependent failures shall be considered as one failure.

Prior to start of this test, the converter shall be run-in for a minimum of 8 hours. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.7 Maintainability. Maintenance operations required during endurance testing as specified in 4.6.2.5.2 and reliability testing as specified in 4.6.2.6 shall be accomplished to determine conformance to 3.8. A demonstrated maintenance ratio greater than the value specified in 3.8.2 and scheduled maintenance greater than that specified in 3.8.3 shall constitute failure of this test. Maintenance support elements such as manuals, lubrication charts, repair parts, and special tools shall be evaluated. Inability of the converter to be properly maintained using the support elements shall constitute failure of this test.

4.6.2.8 Audio noise level. The converter shall be tested for audio noise level at the operators control area when it is delivering liquid at the pressure specified in 3.9.1. Audio noise level measurements shall be measured with an octave band analyzer and sound level meter conforming to ANSI Standard S1.11 and S1.4 respectively. The equal response or flat frequency range shall be used when taking noise level readings. Sound pressure in various frequency bands exceeding the values specified in 3.6.3 shall constitute failure of this test.

4.6.2.9 Lifting attachments.

4.6.2.9.1 Test procedure.

(a) Lift the converter and hold in suspension in its normal travel position using slings that converge not more than 19 feet above the lowest extremity of the converter. Determine the forward angles of application for each attachment. Measure the clearance between each sling and the converter.

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(b) Restrain the converter by anchoring the main frame, and subject each attachment to a pull of 2-1/2 times the load it carried when suspended, applied in the direction as determined in (a) above. Hold each load for 90 seconds.
(c) Insert the tines of a forklift into the tube sockets provided in the converter base, lift and move the converter 5 times. Failure of the converter to remain on the lift tines or damage or deformation to the converter or equipment constitutes failure of this test.

4.6.2.9.2 Failure criteria. Weld failure or permanent deformation of the converter or its lifting attachments, clearance of less than 1 inch between the slings and the converter, or slings converging more than 19 feet above the lowest extremity of the converter shall constitute failure of this test.

4.6.2.10 Tiedown attachments.

4.6.2.10.1 Dynamic. Anchor the converter on a flatcar using the tiedown attachments and any blocking required to prevent movement of the converter when on the railcar. Hump the car, with the converter tied down, at a speed of not less than 7 mph against five cars with their brakes set. The converter shall face in the direction of the five cars. Repeat the procedure with the rear of the converter in the direction of the five cars.

4.6.2.10.2 Static. With the converter anchored by tiedowns other than the attachment being tested, apply static loads as specified to each attachment for a minimum of 90 seconds.

4.6.2.10.3 Failure criteria. Weld failure or permanent deformation of the converter or its tiedown attachments or damage to any tiedown cable shall constitute failure of this test.

4.6.2.11 Environmental.

4.6.2.11.1 High temperature. The converter shall be stored in an ambient temperature of plus 155° F for 4 hours, then examined for damage. The converter shall then be stabilized at plus 120° F. While at this temperature and in an equal ambient temperature, the converter shall be operated at the specified capacity for 2 hours. The cylinder charging operation shall be accomplished in the area of the converter and under the same operating conditions. Malfunction of gages or instruments, inaccessibility of controls, or performance not in accordance with 3.5.2 shall constitute failure of this test.

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4.6.2.11.2 Low temperature. The converter shall be stored in an ambient temperature of minus 65° F for 12 hours then examined for damage. The converter shall then be stabilized at minus 25° F. While at this temperature, and in an equal ambient temperature, the converter shall be started and operated at the specified capacity for 2 hours. The cylinder charging operation shall be accomplished in the area of the converter and under the same operating conditions. Malfunction of gages, controls, or instruments, inaccessibility of controls, or performance not in accordance with 3.5.2 shall constitute failure of this test.

4.6.2.11.3 Rain. The converter shall be subjected to a simulated rainstorm for 1 hour by spraying fresh water at a minimum rate of 3 inches per hour directed simultaneously at the top and at all four sides at an angle of 45 degrees from the vertical. The apparatus for producing the spray shall emit the water in the form of droplets, rather than as a fine mist. The ambient temperature during this test shall be between plus 40° F and plus 90° F. Evidence of any of the following shall constitute failure of this test:

(a) Leakage of water into cabinet.
(b) Inability to start.
(c) Malfunction or inability to operate at full capacity and pressure.
(d) Incorrect operation of any gage, instrument, or control.

4.6.2.12 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.13. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.2.13 Safety. Specific tests and examinations will determine compliance with the requirements of 3.6. In addition, safety aspects of the converter will be evaluated through testing. Failure to comply with 3.6 shall constitute failure of this test.

4.6.2.14 Operational. Operate the converter for a period of 1 hour, using any one of the three gases specified, and test for capacity and performance as specified in 3.9.1 and 3.9.2. All controls shall be operated during testing. All liquid transfer lines shall be tested for conformance to 3.11.6. All components of the system shall function as specified without leakage of vapor or liquid. Inability of the converter to operate for 1 hour at the capacity and pressure specified or malfunction of any component shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select converters at any time during the contract production period and subject these converters to the examination specified in 4.6.1 and to the tests specified in 4.6.2.1 through 4.6.2.14 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Converters will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production converters.

4.7.1 Inspection failure. Failure of an inspection comparison converter to meet any requirement specified herein during and as a result of examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison converter and shall be cause for refusal by the Government to continue acceptance of production converters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on converters previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all converters accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined as follows.

4.8.1.1.1 Components, assemblies, repair parts, and tools that will contact liquid or gaseous oxygen. Examination shall be in accordance with the quality assurance provisions of MIL-C-52211.

4.8.1.1.2 Components, assemblies, repair parts, and tools that will not contact liquid or gaseous oxygen and crated converter. Components, assemblies, repair parts, tools, and crated converter shall be examined for the defects listed in 4.8.2.2.3. Presence of one or more defects shall be cause for rejection of the preproduction pack.

4.8.1.2 Tests. The preproduction pack shall be tested as follows.

4.8.1.2.1 Components, assemblies, repair parts, and tools that will contact liquid or gaseous oxygen. Tests shall be in accordance with the quality assurance provisions of MIL-C-52211.

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4.8.1.2.2 Crated converter. Test shall be in accordance with the quality assurance provisions of MIL-C-52211.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Components, assemblies, repair parts, and tools that will contact liquid or gaseous oxygen. Examination and tests shall be in accordance with the quality assurance provisions of MIL-C-52211.

4.8.2.2 Components, assemblies, repair parts, and tools that will not contact liquid or gaseous oxygen and completed packs of crated converters.

4.8.2.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.2.3 Examination. Samples selected in accordance with 4.8.2.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

157. Electric motors not preserved as specified.
158. Reduction gears not filled with operational lubricant as specified.
159. Repair parts and tools not preserved as specified.
160. Consolidation of components, assemblies, repair parts, tools, and publications not as specified.
161. Materials, method, or containers not as specified. Each incorrect material, method, or container shall constitute one defect.
162. Shroud not furnished for Level B packing as specified.
163. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging, that the applied packaging complies with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government

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(see 6.7). The preproduction pack may be prepared utilizing either the preproduction model or a production converter. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production converters.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Components, assemblies, repair parts, and tools that will contact liquid or gaseous oxygen. Components, assemblies, repair parts, and tools that will contact liquid or gaseous oxygen shall be preserved and unit packed in accordance with MIL-C-52211 and as follows.

5.2.1.1.1 Pump. The pump shall be preserved and unit packed Method A. All sections of piping which present areas subject to possible airlocks, such as those dead-ending at pressure gages, pop safety valves and venting or discharge valves, shall be successively vented until the purging nitrogen exits, and then shall be closed. All bypass valves shall be opened to prevent airlock on one side of the valve. When purging is completed, the pump shall be pressurized to 1.0 psig plus or minus 0.5 psig. A fully-charged cylinder of dry nitrogen shall be furnished; placed and secured on the skid in an accessible location, and shall be tagged indicating: "Pump is pressurized with dry nitrogen". The tag shall conform to UU-T-81, Type A and marking shall be with waterproof ink.

5.2.1.1.2 Hose and fittings, vaporizer, charging manifolds, and charging leads. These components and assemblies shall be preserved and unit packed Method A. Valves on the charging manifolds shall be closed finger-tight. The charging leads shall be bundled together in sets.

5.2.1.1.3 Openings. All drains, overflows, and other openings shall be capped or plugged with fittings provided, or with caps or plugs as specified in MIL-C-52211 when fittings are not provided. Fittings shall be cleaned for oxygen use.

5.2.1.2 Components, assemblies, repair parts, and tools that will not contact liquid or gaseous oxygen.

5.2.1.2.1 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298 by the alternate method.

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5.2.1.2.2 Reduction gears. The reduction-gear housing shall be filled to the operating level with the lubricant recommended for operation and the gears shall be actuated to insure complete coverage of the gears with the lubricant.

5.2.1.2.3 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.2.4 Tools. Handtools shall be preserved in accordance with PPP-P-40.

5.2.1.2.5 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2.1.3 Consolidation. Components, assemblies, repair parts, tools, and technical publications shall be placed in the special storage space provided. Items not provided a special storage space shall be consolidated in the toolbox as space permits. When space is not available for all loose items the remaining items shall be consolidated in close-fitting boxes conforming to PPP-B-601, Domestic Type, style optional. Strapping shall not be required. The boxes shall be secured on the skid in a manner that will prevent damage to the contents or to the converter.

5.2.2 Commercial. Each complete converter with repair parts, tools, and technical publications shall be preserved in accordance with the contractor's standard practice, except that all components, assemblies, repair parts, and tools that will contact oxygen shall be preserved and unit packed in accordance with MIL-C-52211.

5.3 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.3.1 Level A. Each complete converter with repair parts, tools, and technical publications shall be packed in a crate conforming to MIL-C-104, Type I, Class 1 or 2, Style a. Blocking, bracing, and anchoring shall be in accordance with the crate specification and MIL-STD-1186.

5.3.2 Level B. Each complete converter with repair parts, tools, and technical publications shall be packed in a crate conforming to MIL-C-3774, Type I, Style A. The contents shall be protected with a full shroud. Blocking, bracing, and anchoring shall be in accordance with the crate specification and MIL-STD-1186.

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5.3.3 Commercial. Each complete converter with repair parts, tools, and technical publications shall be packed to assure carrier acceptance and delivery to destination, without damage, at the lowest rating of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall reflect, as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible and shall provide a definite contrast.

5.4.3 Special marking. Each preserved and unit packed item processed for oxygen service shall be marked in accordance with the special marking requirements of MIL-C-52211.

6. NOTES

6.1 Intended use. The converter is intended to be used for conversion of liquid argon to gaseous argon, liquid nitrogen to gaseous nitrogen, or liquid oxygen to gaseous oxygen.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) When initial production inspection is required and number of converters to be furnished for initial production inspection, when applicable (see 3.3).
(d) When treatment and painting other than as specified is required (see 3.21).
(e) Technical publications required (see 3.22).
(f) Applicable list of repair parts and maintenance tools required (see 3.23).
(g) When a preproduction pack is not required (see 5.1).
(h) Time frame required for submission of the preproduction pack (see 5.1).
(i) Degree of preservation and degree of packing required (see 5.2 and 5.3).

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6.3 Preproduction model. Any changes or deviations of production converters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish converters conforming to this specification.

6.4 Standard conditions. Standard conditions are defined as 14.7 psi atmospheric pressure and 70° F.

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the converter covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.25.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the converters in accordance with this specification.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair and maintenance tools as necessary, including special tools and instructions on shipment of the converter (see 3.24).

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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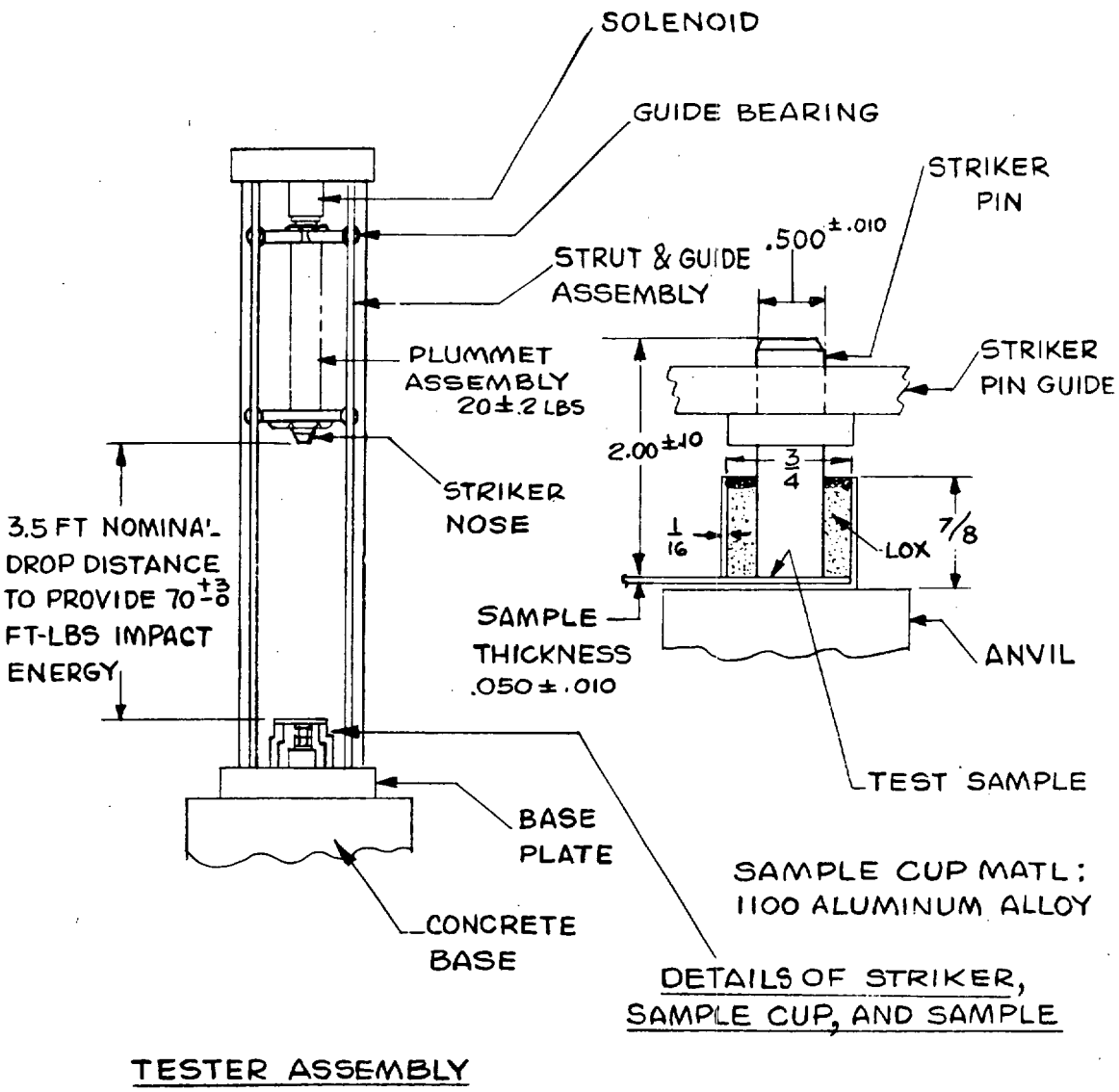


FIGURE 1. Impact sensitivity tester assembly.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3495E(ME) Converter, Liquid- To Gas- Oxygen, Argon, and Nitorgen; Skid-Mounted, 1,200 SCF Per Hour Capacity

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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